AMA572 · Assignment 1 — Complete Solutions
Deadline: September 25, 2026

Q1. Credit Rating Transition Matrices

Use the state order (A, B, C, D), where D is the absorbing default state. Following the corresponding lecture example, interpret "ten months later" as **month 10 measured from the beginning of the year**.

1. Empirical estimator

Count each firm's transition from its initial rating to its year-end rating. The denominator is the number initially in that rating, not the year-end population. The initially B-rated firms that move to A or C remain there for the rest of the year.

Initial rating	Final A	Final B	Final C	Final D	Initial total
A	17	3	0	0	20
B	2	16	2	0	20
C	0	2	15	3	20

The empirical estimate is therefore

$$\boxed{\widehat{P}_{\text{emp}} = \begin{pmatrix} 0.85 & 0.15 & 0 & 0 \\ 0.10 & 0.80 & 0.10 & 0 \\ 0 & 0.10 & 0.75 & 0.15 \\ 0 & 0 & 0 & 1 \end{pmatrix}.}$$

The last row follows from the assumption that default is absorbing.

2. Total time spent in each rating

Let N_i be the total time spent in rating i by all firms, measured in **firm-years**. Include the time spent after entry by firms that migrate into the rating.

$$\begin{aligned} N_A &= 17 \times 1 + 3 \times \frac{3}{12} + 2 \times \frac{11}{12} = \frac{235}{12}, \\ N_B &= 16 \times 1 + 4 \times \frac{1}{12} + 3 \times \frac{9}{12} + 2 \times \frac{5}{12} = \frac{233}{12}, \\ N_C &= 15 \times 1 + 3 \times \frac{7}{12} + 2 \times \frac{10}{12} + 2 \times \frac{11}{12} = \frac{243}{12}. \end{aligned}$$

For example, N_C includes 15 firms that remain in C all year; three that leave C at month 7 (two move to B and one defaults); two that default at month 10; and two that enter C from B at month 1 and remain for 11 months.

3. Generator-based estimator

Let M_{ij} denote the number of observed transitions from i to j. The nonzero counts are

$$M_{AB} = 3, \quad M_{BA} = 2, \quad M_{BC} = 2, \quad M_{CB} = 2, \quad M_{CD} = 3.$$

For $i \neq j$, estimate the transition intensity by

$$\hat{\lambda}_{ij} = \frac{M_{ij}}{N_i}, \qquad \widehat{\Lambda}_{ii} = -\sum_{j \neq i} \hat{\lambda}_{ij}.$$

The estimated generator, in annual units, is

$$\boxed{\widehat{\Lambda} = \begin{pmatrix} -\frac{36}{235} & \frac{36}{235} & 0 & 0 \\ \frac{24}{233} & -\frac{48}{233} & \frac{24}{233} & 0 \\ 0 & \frac{24}{243} & -\frac{60}{243} & \frac{36}{243} \\ 0 & 0 & 0 & 0 \end{pmatrix}}.$$

Numerically,

$$\widehat{\Lambda} \approx \begin{pmatrix} -0.15319149 & 0.15319149 & 0 & 0 \\ 0.10300429 & -0.20600858 & 0.10300429 & 0 \\ 0 & 0.09876543 & -0.24691358 & 0.14814815 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The one-year transition matrix is the **matrix exponential** $\exp(\widehat{\Lambda} \times 1)$, not the elementwise exponential. Evaluating it using the exact fractions gives

$$\boxed{\widehat{P}_{\text{gen}} \approx \begin{pmatrix} 0.86463112 & 0.12857177 & 0.00646155 & 0.00033556 \\ 0.08645026 & 0.82446825 & 0.08249711 & 0.00658438 \\ 0.00416588 & 0.07910217 & 0.78524551 & 0.13148645 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

Checks. Each generator row sums to zero. The transition matrix is nonnegative, its rows sum to one up to rounding, and default remains absorbing. The generator model allows multiple transitions within a year, so it can assign a positive A-to-C probability even without an observed direct A-to-C transition.

As a further check, the total time spent in default is

$$N_D = \frac{5}{12} + 2 \times \frac{2}{12} = \frac{9}{12}, \qquad N_A + N_B + N_C + N_D = 60 \text{ firm-years}.$$

Q2. Piecewise Implied Default Parameters

Use the **linear expected-loss approximation** in Lecture 1: match the price difference between a risk-free bond and a corporate bond to weighted discounted default losses. Let P_1 apply to years 1–3 and P_2 to years 4–5. These are annual default weights under the lecture convention, not exact continuous-time hazard rates.

Take face value 100, annual coupon $c = 4.1$, recovery amount $R = 40$, and continuously compounded risk-free rate $r = 0.025$. Scaling all cash flows by the same factor does not change the estimated parameters.

1. Three-year bond: estimate P_1

Using the quoted continuously compounded yield, the corporate bond price is

$$\begin{aligned} B_3 &= 4.1e^{-0.046} + 4.1e^{-0.046\times 2} + 104.1e^{-0.046\times 3} \\ &= 98.33667697. \end{aligned}$$

The price of the risk-free bond with identical promised cash flows is

$$\begin{aligned} B_3^{\text{rf}} &= 4.1e^{-0.025} + 4.1e^{-0.025\times 2} + 104.1e^{-0.025\times 3} \\ &= 104.47690821. \end{aligned}$$

Let $V_n^{\text{rf}}(t)$ be the time-t risk-free value of the remaining promised payments of the n-year bond. The default loss discounted to time zero is

$$L_n(t) = \big[V_n^{\text{rf}}(t) - 40\big]e^{-0.025t}.$$

Default time t	$V_3^{\text{rf}}(t)$	Loss	Discount factor	$L_3(t)$
0.5	105.791066	65.791066	0.987578	64.973796
1.5	104.317608	64.317608	0.963194	61.950361
2.5	102.806849	62.806849	0.939413	59.001574

For example, at $t = 0.5$, the remaining payments are due at years 1, 2, and 3:

$$V_3^{\text{rf}}(0.5) = 4.1e^{-0.025\times 0.5} + 4.1e^{-0.025\times 1.5} + 104.1e^{-0.025\times 2.5}.$$

Matching the price difference to expected losses gives

$$104.47690821 - 98.33667697 = 185.92573146P_1.$$

Hence,

$$\boxed{P_1 = 0.03302518263 \approx 3.3025\% \text{ per year.}}$$

2. Five-year bond: estimate P_2 for years 4–5

The corporate and risk-free bond prices are

$$B_5 = 4.1\sum_{k=1}^{4} e^{-0.0475k} + 104.1e^{-0.0475\times 5} = 96.67716974,$$

$$B_5^{\text{rf}} = 4.1\sum_{k=1}^{4} e^{-0.025k} + 104.1e^{-0.025\times 5} = 107.28032055.$$

The discounted losses at the possible default dates are as follows.

Time t	$V_5^{\text{rf}}(t)$	Loss	Discount factor	$L_5(t)$	Weight
0.5	108.629741	68.629741	0.987578	67.777209	P_1
1.5	107.228144	67.228144	0.963194	64.753773	P_1
2.5	105.791066	65.791066	0.939413	61.804987	P_1
3.5	104.317608	64.317608	0.916219	58.929006	P_2
4.5	102.806849	62.806849	0.893597	56.124034	P_2

The sum of discounted loss coefficients for years 1–3 is

$$67.77720853 + 64.75377320 + 61.80498675 = 194.33596848.$$

For years 4–5, the sum is

$$58.92900611 + 56.12403367 = 115.05303978.$$

The five-year valuation must use the previously estimated P_1 together with the unknown P_2:

$$107.28032055 - 96.67716974 = 194.33596848P_1 + 115.05303978P_2.$$

Substituting $P_1 = 0.03302518263$ gives

$$\begin{aligned} P_2 &= \frac{10.60315080 - 194.33596848 \times 0.03302518263}{115.05303978} \\ &= 0.03637600501. \end{aligned}$$

Thus, under the lecture's linear approximation,

$$\boxed{P_1 \approx 3.3025\% \text{ per year}, \qquad P_2 \approx 3.6376\% \text{ per year}.}$$

Check and interpretation. Substituting the unrounded estimates reproduces expected losses of 6.14023124 and 10.60315080, equal to the respective bond price differences. Since P_2 applies only to years 4–5, it cannot be obtained by dividing the five-year price difference by the sum of all five loss coefficients.

Q3. Tax Credit and Bankruptcy Cost under GBM

Equations (1) and (2) on page 4 of Lecture 2 are

$$A(0) = E^*\left[e^{-rT}\kappa D\,\mathbf{1}_{\{V_T \geq D\}}\right],$$

$$C(0) = E^*\left[e^{-rT}(1-\alpha)V_T\,\mathbf{1}_{\{V_T < D\}}\right].$$

Here D is the face value of debt, κ is the tax-credit coefficient, and α is the fraction of asset value retained upon default. Under the risk-neutral measure, the assignment appendix specifies

$$dV_t = rV_t\,dt + \sigma V_t\,dW_t, \qquad V_0 = 1.$$

Assume $T > 0$, $D > 0$, and $\sigma > 0$. Let Φ denote the standard normal cumulative distribution function.

1. Distribution of $\ln V_T$

For $f(v) = \ln v$, we have $f'(v) = 1/v$ and $f''(v) = -1/v^2$. Itô's lemma gives

$$\begin{aligned} d\ln V_t &= \frac{1}{V_t}(rV_t\,dt + \sigma V_t\,dW_t) + \frac{1}{2}\left(-\frac{1}{V_t^2}\right)\sigma^2 V_t^2\,dt \\ &= \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW_t. \end{aligned}$$

Integrating and using $V_0 = 1$ yields

$$\ln V_T = \left(r - \frac{1}{2}\sigma^2\right)T + \sigma W_T.$$

Since $W_T \sim N(0,T)$,

$$\ln V_T \sim N\left(\left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right).$$

2. Tax credit $A(0)$

Define

$$d_2 = \frac{-\ln D + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_1 = d_2 + \sigma\sqrt{T}.$$

Using $E^*[\mathbf{1}_E] = P^*(E)$ and

$$P^*(V_T < D) = \Phi\left(\frac{\ln D - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) = \Phi(-d_2),$$

we obtain

$$\boxed{A(0) = \kappa D e^{-rT}\Phi(d_2).}$$

This is the tax credit multiplied by the probability of survival to maturity, discounted to time zero.

3. Bankruptcy cost $C(0)$

Bankruptcy cost depends on asset value at default. We therefore need the **truncated expectation** $E^*[V_T \mathbf{1}_{\{V_T<D\}}]$; the random variable V_T cannot simply be taken outside the expectation.

Write

$$m = \left(r - \frac{1}{2}\sigma^2\right)T, \qquad s = \sigma\sqrt{T}, \qquad \ln V_T = m + sZ, \quad Z \sim N(0,1).$$

The default event is equivalent to $Z < h$, where $h = (\ln D - m)/s = -d_2$. Thus,

$$\begin{aligned} E^*[V_T \mathbf{1}_{\{V_T<D\}}] &= \int_{-\infty}^{h} e^{m+sz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz \\ &= e^{m+s^2/2} \int_{-\infty}^{h} \frac{1}{\sqrt{2\pi}} e^{-(z-s)^2/2}\, dz \\ &= e^{m+s^2/2}\Phi(h-s) \\ &= e^{rT}\Phi(-d_1). \end{aligned}$$

The second equality follows by completing the square:

$$m + sz - \frac{z^2}{2} = m + \frac{s^2}{2} - \frac{(z-s)^2}{2}.$$

Consequently,

$$\boxed{C(0) = e^{-rT}(1-\alpha)e^{rT}\Phi(-d_1) = (1-\alpha)\Phi(-d_1).}$$

Final results for the specified initial value $V_0 = 1$:

$$\boxed{\begin{aligned} A(0) &= \kappa D e^{-rT}\Phi\left(\frac{-\ln D + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right), \\ C(0) &= (1-\alpha)\Phi\left(\frac{\ln D - (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right). \end{aligned}}$$

For a general initial value $V_0 > 0$, the corresponding formulas are

$$d_{1,2} = \frac{\ln(V_0/D) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad A(0) = \kappa D e^{-rT}\Phi(d_2), \quad C(0) = (1-\alpha)V_0\Phi(-d_1).$$

Use the plus sign for d_1 and the minus sign for d_2. Since the lognormal distribution is continuous, $P^*(V_T = D) = 0$; strict and non-strict inequalities therefore give the same expectations here.

Sources: Lecture 1, pp. 6–9 and 13–14; Lecture 2, p. 4, and both supplementary derivations. All inputs follow Assignment 1. The companion *Conventions and Errata* explains the approximation in Q2 and the relevant typographical issues.